COLISTIN

Last-resort antibiotic

Nephro- & neurotoxic

CHEMICAL NAME Polymyxin E (mixture of colistin A and B cyclic polypeptides); given as colistimethate sodium

CLASS Polymyxin antibiotic (last-resort)

TRADE NAMES Colomycin · Coly-Mycin M · Promixin (inhaled) · Colistimethate sodium

THERAPEUTIC USES / INDICATIONS

- Carbapenem-resistant Gram-negative infections: *Acinetobacter baumannii*, *Pseudomonas*, *Klebsiella*.
- Ventilator-associated pneumonia (IV ± nebulised).
- Cystic fibrosis *Pseudomonas* colonisation (nebulised).
- Selective digestive decontamination (oral, ICU).

ADMINISTRATION

- **IV** infusion over 30–60 min (or slow injection via central line).
- **Nebulised** — give bronchodilator before (bronchospasm risk).
- Dose units: 1 million IU ≈ 80 mg colistimethate ≈ 33 mg colistin base activity — check carefully.

DOSAGES

- **Adult IV:** loading 9 million IU, then 9 million IU/day in 2–3 divided doses (normal renal function).
- **Nebulised:** 1–2 million IU every 8–12 h.
- **Children IV:** 75,000–150,000 IU/kg/day divided.
- Reduce maintenance dose in renal impairment.

PHARMACOKINETICS

- Colistimethate (prodrug) hydrolysed to active colistin.
- Poor penetration into lung, CSF.
- Excreted mainly in urine. **t½:** 4–9 h (prolonged in renal failure).

PHARMACODYNAMICS

- Cationic detergent: binds **lipopolysaccharide** of Gram-negative outer membrane → disrupts membrane → bactericidal.
- Concentration-dependent killing; no Gram-positive activity; *Proteus*, *Serratia* resistant.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Perioral and limb paraesthesia, dizziness.
- Injection-site pain.
- Cough, bronchospasm (inhaled).

ADVERSE EFFECTS (SERIOUS)

- Nephrotoxicity (acute tubular necrosis) in up to 30–60%.
- Neurotoxicity: neuromuscular blockade → apnoea, confusion, vertigo.
- Bronchospasm with nebulisation.
- Rare: anaphylaxis.

CONTRAINDICATIONS & CAUTIONS

- Polymyxin hypersensitivity.
- Myasthenia gravis.
- **Caution:** renal impairment, use with other nephrotoxic or neuromuscular-blocking drugs.

KEY INTERACTIONS

- **Aminoglycosides, vancomycin, amphotericin:** ↑ nephrotoxicity.
- **Neuromuscular blockers, anaesthetics:** apnoea.

NURSING MANAGEMENT

- Double-check dose units (IU vs mg).
- Monitor creatinine and urine output daily.
- Watch for respiratory muscle weakness, apnoea.
- Ask about tingling, dizziness.
- Inhaled: bronchodilator first; monitor SpO_2 and wheeze.

PATIENT EDUCATION

- Report numbness, tingling, dizziness, weakness or less urine.
- Use nebuliser technique as taught; rinse mouth after.

KEY POINTS

- Polymyxin; last resort for carbapenem-resistant Gram-negatives.
- Confusing units — check IU vs mg.
- Nephrotoxic and neurotoxic.
- No Gram-positive cover.